Graph Patterns and Size in Knowledge-Graph Entity Selection: Evidence-Conditional Profiles of Large Language Models

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Abstract

Selecting entities that support a knowledge-graph query requires recovering intermediate context as well as plausible endpoints. We analyze archived evaluations of three language models on seven graph-pattern task families with 25 candidates. A controlled panel compares families at common or explicitly distinguished sizes; a second panel covers 34 motif-size cells per model under native entity cards and an explicit relation overlay. At six nodes, exact recovery for outward stars is 1.00, 0.99, and 0.99 for Gemma, Qwen, and Ministral, compared with 0.87, 0.15, and 0.54 for inward trees. Size profiles depend on family and model: several chains and trees lose quality, while outward-star profiles can remain near ceiling. Adding relations increases mean graph-reference F1 by 0.674, 0.466, and 0.572, respectively; adjacent distractors still reduce selection quality. We distinguish route coverage from minimum sufficient evidence and answer correctness, retain invalid outcomes in planned denominators, and qualify conditional uncertainty and historical evidence limits. The findings support evaluating structural profiles under explicit evidence conditions rather than a universal node-count or topology difficulty law.

CCS Concepts

• **Information systems** → **Information retrieval**; • **Computing methodologies** → *Natural language processing*.

Keywords

knowledge graphs, entity selection, graph patterns, language models, evaluation

ACM Reference Format:

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1 Introduction

A question about a company, a product, and a news event can require several linked entities even when the final answer is a single name. In knowledge-based access to Web information, an intermediate selection step must retain these entities before an answer generator can use their relations. Selecting plausible endpoint names is insufficient when intermediate evidence is omitted; selecting many connected names can instead admit irrelevant neighbours. Understanding these errors requires examining the structure of the requested evidence, not only average selection accuracy.

We study this selection step on a fixed candidate pool drawn from a corporate knowledge graph (KG). The resource connects companies, people, products, locations, and news and was developed for corporate GraphRAG evaluation [6]. Our Web-related question is how reliably language models select the connected entity context needed to access and combine this structured information. We evaluate agreement with an explicitly defined entity-route reference, not the performance of a deployed Web search system or the correctness of generated answers.

Graph encodings and graph-pattern comprehension are established subjects of LLM evaluation [4, 5]. The closely related Tool Graph Complexity manuscript already includes KG entity selection, motif and size comparisons, and adjacent distractors in Appendix O [1]. We take that KG line as a related experimental foundation and examine structural profiles under distinguished evidence and presentation conditions. We do not claim that motif-based KG selection, size effects, or distractor sensitivity are new phenomena. The precise extent of shared historical task/response records with that appendix has not been independently established, and we make no independent-replication or new-resource claim.

Our analysis has three questions. **RQ1:** At comparable node counts, which graph-pattern task families are difficult for each model, and what selection errors occur? **RQ2:** Within a family, how does quality vary across available sizes and evidence conditions? **RQ3:** How robust are these profiles to explicit relation availability and adjacent distractors? Rendering, card placement, wording, and naming provide supplementary diagnostics rather than separate headline research questions.

The study makes three scoped contributions: (i) a descriptive characterization of pattern-family profiles, separating equal-size comparisons from families evaluated at different sizes; (ii) an all-model analysis of family-specific size profiles under native cards and their explicit relation overlay; and (iii) a reference-aware account of condition sensitivity, partial versus exact recovery, and omission versus inclusion errors. Audited pairing and failure-inclusive reporting make these comparisons inspectable. These are empirical contributions on a fixed KG, not a new selection algorithm.

The principal observations resist a single complexity ordering. At six nodes, outward stars are recovered much more often than inward trees, especially for Qwen. Increasing size is associated with substantial losses for some chains and trees but almost no loss for outward stars in some model/condition combinations. Explicit relations substantially increase recovery of the graph reference, while adjacent contexts remain difficult. Since motif families also differ in query semantics and accepted task composition, these results are conditional profiles rather than isolated causal effects of topology.

2 Related Work

Graph structure and representation. Talk like a Graph investigates graph-to-text encodings and variation across graph tasks and

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structures [5]. Dai et al. evaluate terminology-based and topology-based pattern comprehension and pattern discovery [4]. Our output is an entity set covering query-consistent routes in a supplied candidate context; it is not a pattern name, a discovered motif, or an edited graph. Choosing motifs alone therefore does not distinguish the study from this literature.

Knowledge-graph prompts. Dai et al. study KG input formats, triple organization, and robustness to noisy or incomplete subgraphs, including answer-based evaluation [3]. We similarly treat format and noise as relevant factors, but separate graph-reference recovery from answer correctness. Our templated prose is a controlled expression of designated facts, not arbitrary fluent natural language.

Tool Graph Complexity and its KG substrate. The supplied manuscript distinguishes unnecessary reference tools from the way required tools compose [1]. Its Appendix O uses a finance KG, 25-name pools, explicit relations, motif/size variation, and adjacent decoys. It reports partial transfer of tool-graph rankings, with chains easier on explicitly represented KG relations and incoming trees difficult. Its selection-completeness metric does not penalize extra selections, whereas exact recovery does. These are direct points of overlap, not merely an analogy between APIs and entities.

Our focus is the conditional KG profile across native evidence, explicit overlays, and controlled contexts. The additional analyses address how structural profiles should be interpreted when information availability and presentation differ. We do not compare the two papers' scores as a leaderboard: their populations, reference conventions, and metrics are not harmonized. Graph construction and the earlier corporate QA benchmark belong to Kirpichev et al. [6]; they are not presented as new resources here.

3 Task and Experimental Material

Source graph and historical scope. We reuse the corporate KG snapshot described by Kirpichev et al. [6], covering S&P 500 companies and associated sectors, industries, executives, investors, products, resources, locations, and news. The authors confirm unchanged snapshot reuse. We analyze the archived, frozen tasks rather than reconstruct their historical database extraction from current software. Appendix B reproduces the author-supplied Cypher extraction specification. It defines fixed small instances and an induced-edge filter; it is not presented as the complete executable pipeline for all archived sizes or per-cell sampling quotas. No database query was executed during this analysis.

Task families and questions. The Core panel has 100 base tasks in each of seven families: chain, star-in, star-out, tree-in, tree-out, rhombus, and hourglass. Five families have six entities; rhombi have four and hourglasses five. Chain questions follow outgoing routes until the first entity of a specified type on each route. Stars ask for incoming or outgoing one-hop neighbours. Trees ask for two-step incoming or outgoing traversal; rhombi use two-step outgoing traversal. A two-step route terminates early only if no further unvisited neighbour is permitted. Hourglass questions request incoming and outgoing one-hop neighbours in separate groups.

Question templates are instantiated using the anchor name/-type, direction, and stopping rule. Step-by-step and direct formulations express the same endpoint query; the former enumerates the traversal operations. These families thus vary in topology *and* query operation. In particular, outward trees and rhombi share the outgoing two-step query, while other families use different directions or stopping rules. An unadjusted seven-family ranking is neither an equal-size nor a pure-topology comparison.

Candidates and distractors. Each main task presents 25 named candidate entities, including the motif entities and distractors. Disconnected distractors have no edges to reference entities in either direction, although they may connect to each other. In the adjacent variant, up to ten distractors are neighbours of motif entities; remaining distractors are disconnected. Paired noise variants retain the question and reference but may contain different distractor identities. Explicit evidence includes relations involving distractors, not only reference-induced edges.

Selection target. Let C denote the candidate catalogue and G_C its graph-defined reference context, which need not be recoverable from native cards. Let q be the endpoint query, a its anchor, and $\mathcal{P}(q, G_C)$ its allowed supporting routes under the saved traversal/stopping rule. We evaluate the entity-route reference

$$W(q, G_C) = \{a\} \cup \bigcup_{p \in \mathcal{P}(q,G_C)} V(p), \tag{1}$$

where $V(p)$ contains every entity on route p. Endpoint answers $A(q, G_C)$ are distinct from W. Simple routes do not revisit an entity; the saved task specification fixes the allowed direction, depth, and terminal-type rule. In the audited structured archive, saved motif/reference IDs coincide with this witness union. Equality is a property of these accepted tasks, not an assumed identity for arbitrary candidate expansion.

For example, one archived rhombus has two branches from the anchor through two intermediate entities to one endpoint. Its endpoint answer has one entity and its reference four. Removing the endpoint card's facts while retaining the name/type catalogue leaves that endpoint recoverable from incoming facts. Thus reference membership does not establish that every card contributes indispensable information. The selection objective is complete route coverage, not minimum sufficient evidence or final answer correctness.

Evidence conditions. Raw presents original entity metadata; Raw+E adds explicit relations to the same cards. Actual archived inputs, questions, candidate order, and references match after removing only the relation fields. This is an information-availability contrast; native descriptions are not assumed to express all graph relations. Controlled triples and templated prose instead encode designated matched graph facts. Source-card, destination-card, and both-card variants alter presentation location while retaining relation direction. Names/types remain visible in all conditions. Pseudonyms, direct questions, and linked-attribute encodings provide further diagnostic contrasts detailed in Appendix A.

Table 1: Main panels. Counts are bases per condition/model; repeated conditions and overlapping panels are not independent samples to add together. All main pools contain 25 candidates.

Panel	Coverage	Bases
Core	Seven families; 13 conditions; all three models	700
Full Raw/Raw+E	34 motif-size cells; both conditions; all three models	3,400
Controlled size	Extra K1/K5 sizes; Gemma only	1,100

4 Experimental Design and Evaluation

Models and populations. We analyze the archived Gemma-4-31B, Qwen3.6-Flash, and Ministral-14B evaluations. Model labels identify the archive, not guaranteed reproducible present-day endpoints. Core contains 700 base tasks under 13 conditions, or 9,100 planned outcomes per model. Full Raw/Raw+E has 3,400 bases per condition/model. Its size grid is chain 3–6, stars 3–10, trees 5–10, rhombus 4, and hourglass 5: 34 cells, each with 100 tasks. The separate controlled additional-size panel covers 1,100 Gemma bases per K1/K5 condition. It is a supplemental check, not the only evidence about size.

The 700 Core base IDs occur in Full Raw/Raw+E for each model; panels must not be counted as independent replications. More broadly, base IDs recur across conditions and models. Within an evaluated task/model/condition, transport retries are not additional scientific replicates. Table 1 reports planned bases rather than successful calls.

Principal comparisons. For RQ1, we compare Core source-card triples with disconnected (K1) and adjacent (K2) distractors. The first five motif families provide a common six-node comparison; the two remaining families retain their native sizes. RQ2 uses the complete three-model Full Raw/Raw+E size grid, without pooling across families. RQ3 uses audited Raw/overlay pairs and the two Core noise pairs, K1/K2 for triples and K5/K6 for templated prose. Rendering, placement, wording, naming, linked attributes, and additional access settings are supplementary diagnostics.

Responses and evidence tiers. The main protocol requests a JSON object with selected candidate names and an insufficient-evidence flag. Valid selections use exact, unique catalogue names. The flag does not modify set scores. Available Qwen and Ministral request records in the main comparisons specify OpenRouter, temperature zero, a 1,024-token output limit, and JSON-object formatting. We do not impute these settings to unavailable Gemma call records.

The broader archive includes 12,400 Gemma Core/size/calibration/incremental outcomes with frozen inputs and cached individual scores but no original raw replies. Full Raw/Raw+E has stronger raw-response coverage; 15 Ministral Raw outcomes lack a reply after exhausted attempts and remain failures in the planned denominator. We analyze the saved outcomes and preserve these evidence tiers. Offline verification of available requests, selections, and references is distinct from rerunning a provider or authenticating missing historical records.

Metrics. For valid S and a nonempty reference W, precision is $|S \cap W|/|S|$, recall is $|S \cap W|/|W|$, and

$$F_1(S, W) = \frac{2|S \cap W|}{|S| + |W|}. \tag{2}$$

An empty selection receives zero precision, recall, and F1. Exact selection is one only if $S = W$. We report means of individual task scores over all planned outcomes; invalid and missing outcomes contribute zero, with original missing individual metrics retained in the archive. The resulting mean F1 is not pooled micro-F1. Valid-only scores are supplementary because conditioning on successful output changes the population.

For valid outcomes we additionally distinguish omission-only ($\mathrm{FN} > 0, \mathrm{FP} = 0$), inclusion-only ($\mathrm{FP} > 0, \mathrm{FN} = 0$), and mixed errors. Invalid responses remain separate. These observable categories do not identify a model's internal reasoning mechanism.

Uncertainty and descriptive scope. Motif profiles and size curves are descriptive slices of the accepted frozen tasks; they carry no newly fitted topology coefficients, interaction tests, or significance claims. Sizes are different sampled strata, not randomized expansions of one task. For previously audited paired condition contrasts, we use 10,000 anchor-bootstrap replicates, sharing each anchor weight across aligned tasks, models, and conditions and retaining original motif-size weights. Paired task resampling is a sensitivity check. The distinction between within-cluster dependence and independence across clusters is essential [2].

Pointwise intervals are bootstrap percentiles. Exploratory simultaneous bands cover the declared family of 78 F1/exact condition effects: each panel uses a centered maximum standardized deviation at nominal 97.5% coverage, with a two-panel union bound for combined nominal coverage of at least 95%. This is an approximate bootstrap construction, not finite-sample coverage or a collection of p-values. The plan followed prior descriptive inspection. These bands do not cover arbitrary new motif/size comparisons. Cross-anchor shared-entity dependence remains a limitation (Section 6).

5 Results

5.1 RQ1: Pattern-Family Profiles

Figure 1 shows that a single aggregate score conceals large task-family differences. Under K1, six-node star-out exact recovery is 1.00 for Gemma and 0.99 for Qwen and Ministral. Six-node tree-in exact recovery is 0.87, 0.15, and 0.54, respectively. The corresponding tree-in F1 values are 0.984, 0.734, and 0.899. Thus substantial partial overlap can coexist with frequent failure to recover the complete reference. For Qwen, outward-tree exact recovery is 0.50, also below star-out. These comparisons hold node count and candidate-pool size fixed, but not query semantics, relation types, or sampled entities.

The profiles differ across the evaluated models descriptively. Gemma reaches the ceiling for several disconnected families; its remaining difficulty is clearer under exact recovery and adjacent

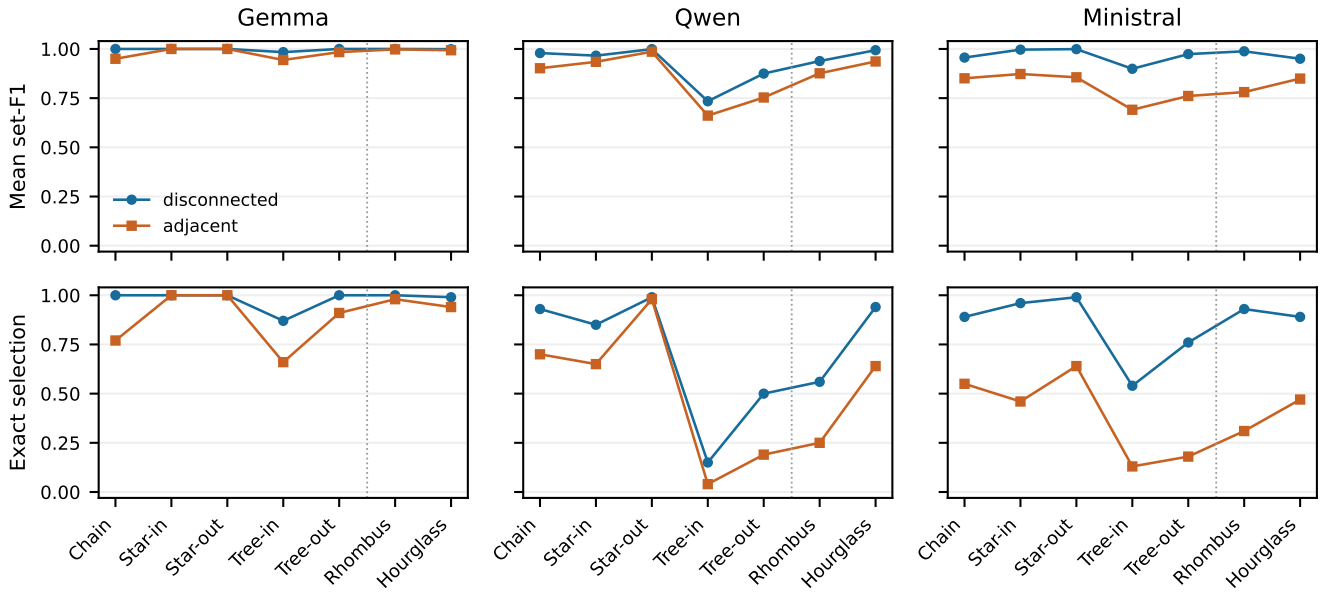


Figure 1: Core pattern profiles with source-card triples: disconnected K1 and adjacent K2. Each cell has 100 planned tasks per model/condition; invalid outcomes contribute zero. The first five families have six nodes. Rhombus (four) and hourglass (five), beyond the dotted separator, are not equal-size comparisons. These are descriptive means without inferential error bars. F1 and exact recovery reveal different aspects of failure.

contexts. Qwen's tree-in errors under K1 comprise 83 omission-only outcomes, two mixed outcomes, and 15 exact outcomes out of 100. The failure is predominantly incomplete selection rather than wholesale unrelated selection. We do not infer a hidden search procedure from that observation.

Adjacent contexts expose further differences. Ministral's K2 star-out outcomes include 31 inclusion-only, one mixed, four invalid, and 64 exact outcomes. Its K2 tree-in exact recovery is 0.13, compared with 0.54 in K1. Qwen's tree-in exact recovery falls from 0.15 to 0.04. Gemma's star-out remains at 1.00, whereas its chain exact recovery falls from 1.00 to 0.77. The noise response is therefore not adequately summarized as an identical penalty on every family. These slices are descriptive; we have not tested a model-by-motif-by-noise interaction.

Rhombus and hourglass results complete the evaluated family catalogue but should not determine an equal-size ranking. Their smaller references change the amount of required selection, and their queries differ from several six-node families. In particular, a high hourglass score here does not establish that joins or fan-in are universally easy. We use the family labels to identify empirical task profiles, not to claim a task-independent ordering of abstract graphs.

5.2 RQ2: Pattern-Specific Size Sensitivity

The complete Full Raw/Raw+E grid supports size analysis for all three models, not just Gemma (Figure 2). However, the observed curves do not support a universal "more nodes means worse selection" statement. Under Raw+E, Qwen chain F1 changes from 0.994 at three nodes to 0.848 at six, and Ministral from 0.929 to 0.717. Gemma changes from 0.990 to 1.000. Chain length therefore distinguishes the evaluated models in this condition rather than imposing the same observed degradation on each.

Outward stars provide a contrasting profile. Qwen Raw+E F1 is 0.998 at both three and ten nodes after rounding, while Gemma is 1.000 at both endpoints. Ministral increases from 0.922 to 0.973. These endpoint comparisons do not demonstrate invariance or beneficial causal effects of added nodes: ceiling scores, different accepted samples, and reference prevalence all matter. Inward stars are less stable for Qwen, declining from 0.976 to 0.755 across the same range. Directional family distinctions remain relevant even when both families have a hub-and-leaf description.

Incoming trees deteriorate between five and ten nodes in all three Raw+E model profiles: Gemma F1 changes from 0.940 to 0.875, Qwen from 0.758 to 0.530, and Ministral from 0.685 to 0.536. Exact recovery at these endpoints is 0.67 to 0.60, 0.07 to 0.02, and 0.03 to 0.04, respectively. The latter illustrates why partial and exact metrics must be retained: a lower F1 endpoint does not imply a lower exact endpoint in every small stratum. Some intervening curves are nonmonotonic. Outward-tree endpoint values, and the complete set of size endpoints, are included in Appendix Table 5 rather than selecting only decreasing examples.

The size axis simultaneously changes the number of target entities and distractors: reference size s leaves $25 - s$ distractors. Even selecting all candidates has F1 $2s/(25 + s)$, increasing with s. Exact recovery also requires every member of a larger reference to be correct. Consequently, the curves characterize performance on the frozen size grid, not an intrinsic graph-complexity coefficient.

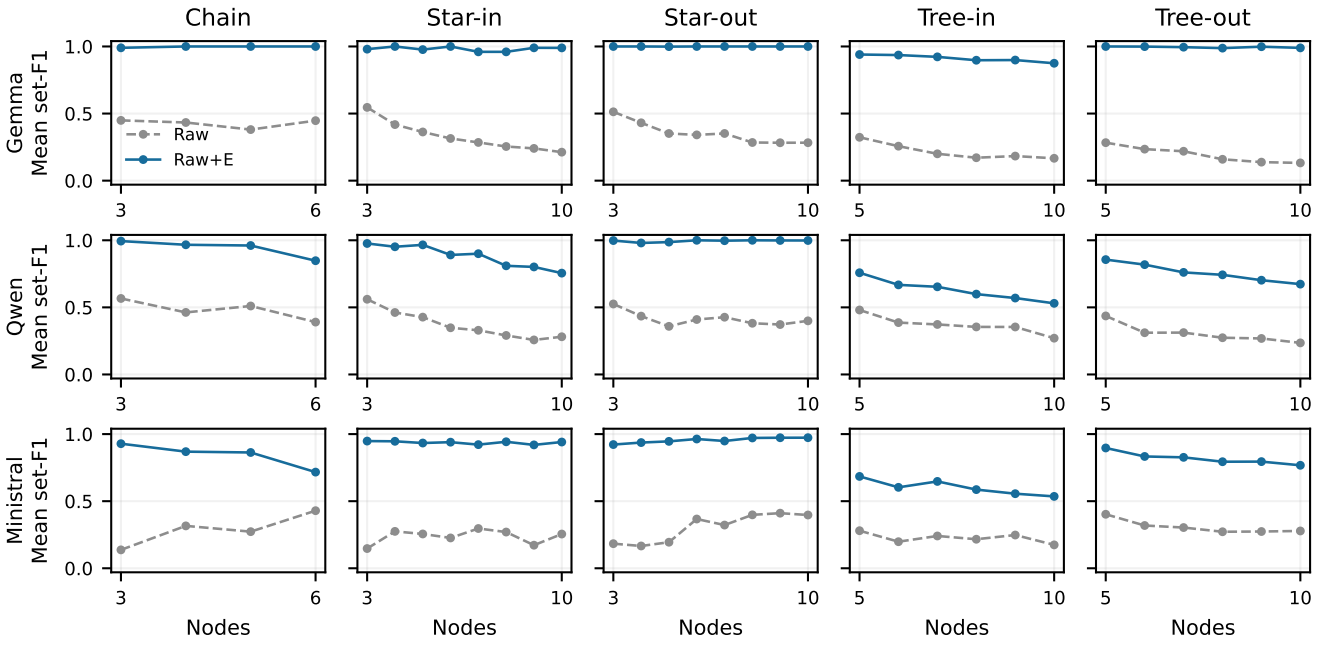


Figure 2: Size profiles for all variable-size families in Full Raw/Raw+E. Rows are models and columns are families; each point averages 100 planned tasks, retaining invalid outcomes as zero. The pool remains at 25 candidates. Grey dashed lines are Raw; blue solid lines are Raw+E. Sizes are different frozen samples, not paired graph expansions. No inferential error bars or universal monotonicity claim are implied. Rhombus and hourglass have one available size and are omitted from size curves. Appendix Table 5 reports endpoint F1 and exact recovery.

Table 2: Paired condition effects in score units. Raw contrast: 3,400 pairs/model; noise contrasts: 700. All planned outcomes retained. Bands are exploratory simultaneous bands from the full 78-effect F1/exact family under the fixed-KG/anchor-independence assumptions; they are not new motif/size tests.

Model	Contrast	ΔF1	Simultaneous band
Gemma	Raw+E − Raw	+0.674	[0.664, 0.683]
Gemma	Adjacent − disc., triples	-0.016	[-0.026, -0.007]
Gemma	Adjacent − disc., text	-0.023	[-0.038, -0.008]
Qwen	Raw+E − Raw	+0.466	[0.453, 0.479]
Qwen	Adjacent − disc., triples	-0.062	[-0.086, -0.038]
Qwen	Adjacent − disc., text	-0.058	[-0.082, -0.033]
Ministral	Raw+E − Raw	+0.572	[0.554, 0.590]
Ministral	Adjacent − disc., triples	-0.157	[-0.191, -0.124]
Ministral	Adjacent − disc., text	-0.166	[-0.198, -0.134]

The additional Gemma K1/K5 panel is a controlled-format diagnostic; it does not supply a full all-model crossing of size, format, and adjacent noise.

5.3 RQ3: Evidence Availability and Adjacent Contexts

Adding explicit relations substantially increases recovery of the graph reference (Table 2). Failure-inclusive mean F1 changes from 0.305 to 0.978 for Gemma, 0.386 to 0.852 for Qwen, and 0.275 to 0.847 for Ministral. Paired gains are +0.674, +0.466, and +0.572; all corresponding simultaneous bands are positive under the stated assumptions. The gain remains positive in the both-valid diagnostic, so it is not explained solely by a greater frequency of invalid Raw outcomes. Raw+E adds information rather than merely reformatting an identical fact set. Low Raw agreement cannot be interpreted as a reasoning deficit unless sufficient native evidence is established independently.

In the controlled Core panel, adjacent-minus-disconnected F1 is negative for every model under both triples and templated prose. The six simultaneous bands lie below zero under the conditional bootstrap analysis. Absolute penalties differ substantially, from approximately 0.016–0.023 for Gemma to 0.157–0.166 for Ministral. Context changes include distractor composition and density, so these are not pure causal estimates of an abstract adjacency property. Pattern profiles in Figure 1 show why aggregate penalties should be read alongside family-level failure modes.

The presentation checks are smaller and model-dependent. Qwen's text-minus-triples F1 differences are about −0.023 under both question formulations, with negative simultaneous bands;

Gemma and Ministral bands include zero. This is not an equivalence finding for those models. Duplicating facts on both cards gives Qwen a +0.041 F1 difference, with simultaneous band [0.024, 0.059], but no uniform benefit across models. Appendix A reports alternative placement/rendering estimates and all Core condition means, preserving unfavourable and small differences as well as favourable ones.

6 Discussion, Limitations, and Reproducibility

What the structural profiles establish. The common-size contrast between outward stars and incoming trees shows that task-family difficulty is not explained by node count alone. Size curves then show that increasing reference size need not produce the same profile across families or evaluated models. A useful evaluation should therefore stratify by family, model, and evidence condition rather than collapse them into a universal graph-difficulty score. This conclusion concerns the measured entity-selection task; it does not rank the general reasoning ability of a model.

The structural axis remains conditional. Incoming/outgoing orientation, traversal depth, terminal-type rules, entity/type distributions, and question wording vary with family. Accepted samples can differ across sizes. Fixed candidate count does not fix target prevalence, context length, graph density, or per-entity selection difficulty. Ceiling effects obscure small losses in strong conditions. The study does not isolate topology from all these factors, and we report no regression-adjusted causal or significance claim for motif or size.

Relation to the KG appendix. The KG experiment in Tool Graph Complexity already studies motif and size effects with explicit relations and adjacent neighbours [1]. We position our work as a focused examination of that related KG line under additional evidence and presentation contrasts. We do not claim independent validation of Appendix O or new construction of its KG substrate. Its exact historical row overlap with this archive is not independently quantified. Scores across the papers should not be directly subtracted or interpreted as changes in a model's capability. The additional value here is the explicit conditional analysis and interpretation, not rediscovery of KG motif or distractor effects.

Reference validity and downstream utility. Reference coverage, minimal graph support, indispensable visible cards, and correct answers are distinct. Matching W demonstrates adherence to the route-coverage convention. A smaller selected set could still contain sufficient evidence for an endpoint answer; a large set can receive nonzero F1 without grounded reasoning. Analytical references illustrate this: select-all F1 is 0.364 on Core and 0.405 on Full Raw, despite not using the question or facts. We provide these as metric reference points, not executed retrieval systems or competitive QA baselines. There is no validated downstream QA improvement in this study.

Sampling, uncertainty, and generalization. One corporate KG and its accepted frozen tasks cannot establish transfer to other domains, snapshots, model versions, or prompts. Anchors repeat and reference entities cross anchor clusters. Shared entities connect 578 of 700 Core tasks and 3,265 of 3,400 Full Raw tasks in their largest reference-overlap components. Anchor bootstrap intervals thus rely on cross-anchor independence conditional on the fixed KG; they do not control arbitrary shared-entity dependence. The nominal simultaneous bands and task-bootstrap sensitivity should be read within that restriction. Transport retries and multiple response stages are not independent repetitions, and we have no generation-seed replication study.

Historical evidence and reproducibility. The archived observation, input, response, and task-specification files retain IDs and hashes, allowing task/reference checks and bounded offline score aggregation. Some Gemma outcomes have cached scores without raw replies; unavailable call settings cannot be inferred from current defaults. Incomplete incremental histories prevent equal-budget or mechanism claims about interactive access. The full source snapshot was not loaded during the audits, so frozen-context agreement does not prove completeness against the original database.

The accompanying analysis materials identify source rows, planned denominators, pairing keys, failure policy, resampling units, and figure provenance. Motif/size plots are derived from existing audited summaries and add no new inference. Controlled and legacy parsing/scoring conventions remain distinct; legacy outcomes are not pooled into the main results. Appendix queries document read-only access/checks without representing their execution as an experiment. Public release of raw resources and the resolution of any access restrictions are separate from the reproducibility of these archived summaries.

Data and writing responsibility. The analysis uses existing corporate-KG experimental artifacts and includes no new human-subject study or new model calls. We do not infer consent, redistribution rights, or an ethics-board decision from the presence of an archive. Generative AI assistance was used to organize the evidence audit, prepare analysis/plotting code, and draft and edit this manuscript; the numerical summaries derive from the saved experimental data rather than invented model outputs. The authors remain responsible for checking the manuscript and its relationship to the related KG study.

7 Conclusion

Knowledge-graph entity selection exhibits task-family profiles that node count alone does not describe. On the archived common-size tasks, incoming trees are markedly less often recovered exactly than outward stars. Across available sizes, several chains and trees lose quality while other profiles remain near ceiling or vary nonmonotonically. Explicit relations greatly increase graph-reference recovery, yet adjacent contexts still expose omission and excess-selection errors. These findings support evaluating structural profiles under clearly specified evidence conditions, with partial and exact recovery reported together. They do not establish a universal topology ranking, indispensable-card minimality, or downstream QA gains.

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A Complete Conditions and Supplementary Results

Condition definitions. Core conditions use pseudonyms, step-by-step questions, and disconnected distractors unless specified. K1 is source-card triples; K2 changes to adjacent distractors; K3 puts triples on destination cards; K4 duplicates them on both cards. K5 is templated text and K6 its adjacent variant. K7/K8 use direct questions with triples or text. K5k encodes links through matching attribute values rather than names. R1-matched and R2-matched use real names with disconnected or adjacent triples. R0/K0 provide real names or pseudonyms and types only. Their prompt permits prior name knowledge and requests an insufficient-evidence flag when relations cannot be established; other controlled conditions restrict evidence to supplied cards.

Representation, wording, placement, and naming pairs retain the relevant candidate/reference/fact invariants. Noise pairs retain the question and reference but deliberately alter distractor context. These are not interchangeable interventions. Table 3 includes small and null differences as well as large ones.

Table 3: All Core conditions: mean F1 / exact selection, each over 700 planned tasks/model. Invalid outcomes contribute zero. These are descriptive absolute means, not confidence intervals.

Condition	Gemma	Qwen	Ministral
K0	0.307 / 0.000	0.308 / 0.000	0.160 / 0.000
K1	0.998 / 0.980	0.926 / 0.703	0.966 / 0.851
K2	0.981 / 0.894	0.864 / 0.493	0.809 / 0.391
K3	0.997 / 0.984	0.930 / 0.689	0.936 / 0.743
K4	0.994 / 0.986	0.968 / 0.844	0.944 / 0.817
K5	0.996 / 0.971	0.903 / 0.630	0.964 / 0.841
K6	0.973 / 0.871	0.845 / 0.421	0.799 / 0.394
K7	0.998 / 0.986	0.924 / 0.679	0.952 / 0.811
K8	0.994 / 0.959	0.901 / 0.607	0.959 / 0.771
K5k	0.991 / 0.960	0.899 / 0.574	0.933 / 0.707
R0	0.437 / 0.034	0.389 / 0.007	0.300 / 0.020
R1-matched	0.988 / 0.960	0.874 / 0.564	0.892 / 0.581
R2-matched	0.975 / 0.884	0.816 / 0.361	0.763 / 0.214

Table 4: Selected controlled presentation contrasts, paired mean F1 differences over 700 planned pairs/model. Their full interval records remain in the analysis archive. Means crossing zero in uncertainty analyses are not evidence of equivalence.

Contrast	Gemma	Qwen	Ministral
Text − triples, steps	-0.002	-0.023	-0.002
Text − triples, direct	-0.004	-0.023	+0.006
Destination − source	-0.001	+0.003	-0.030
Both − source	-0.003	+0.041	-0.023

Table 5: All variable-family Raw+E size endpoints: smallest (s) and largest (l) available sizes, with 100 planned tasks per endpoint/model. These are descriptive comparisons of different sampled strata; endpoint direction does not imply monotonic intermediate values.

Model	Family	Sizes	$F1_s$	$F1_l$	$Exact_s$	$Exact_l$
Gemma	Chain	3–6	0.990	1.000	0.990	1.000
Gemma	Star-in	3–10	0.980	0.989	0.980	0.980
Gemma	Star-out	3–10	1.000	1.000	1.000	1.000
Gemma	Tree-in	5–10	0.940	0.875	0.670	0.600
Gemma	Tree-out	5–10	1.000	0.989	1.000	0.980
Qwen	Chain	3–6	0.994	0.848	0.970	0.630
Qwen	Star-in	3–10	0.976	0.755	0.930	0.570
Qwen	Star-out	3–10	0.998	0.998	0.990	0.970
Qwen	Tree-in	5–10	0.758	0.530	0.070	0.020
Qwen	Tree-out	5–10	0.856	0.673	0.320	0.180
Ministral	Chain	3–6	0.929	0.717	0.900	0.190
Ministral	Star-in	3–10	0.948	0.941	0.740	0.490
Ministral	Star-out	3–10	0.922	0.973	0.610	0.500
Ministral	Tree-in	5–10	0.685	0.536	0.030	0.040
Ministral	Tree-out	5–10	0.897	0.768	0.550	0.130

Analytical reference points. Select-all and anchor-only scores are deterministic given the reference. Uniform random-six samples six of 25 candidates without replacement; its entries are exact expectations, not simulated model outputs. In order select-all, anchor-only, random-six, Core F1 values are 0.364, 0.309, and 0.230; Full Raw F1 values are 0.405, 0.292, and 0.242. These reference points do not establish evidence-grounded retrieval or answer correctness. Their interpretation depends on the reference convention and target prevalence.

Additional archive panels. Gemma has the separate K1/K5 additional-size panel. The matched incremental comparison uses 350 bases per condition, with endpoint F1 differences of −0.237 for triples and −0.232 for text relative to static evaluation on those same subsets. Incomplete reveal histories and unmatched token budgets prevent an equal-budget or mechanism interpretation. Native-description and legacy panels retain their own semantic goals, reference/scoring policies, and populations; they are not pooled with Core or Full Raw. Calibration microtasks are also separate from the main 25-candidate task.

B Author-Supplied Cypher Extraction Specification

The following Cypher query was supplied by the authors for the appendix. Whitespace is normalized and comments are translated into English for display; executable tokens match the saved original. It enumerates small instances of nine directed pattern definitions, requires distinct vertices, and excludes additional directed typed edges inside each selected vertex set, including self-loops. The comparison uses source/type/target triples, so parallel physical relationships with the same triple are not distinguished by the extra-edge test.

This specification has fixed sizes: four-node chains/stars/rhombi, five-node trees/hourglasses, and three-node cycles/cliques. Its final `LIMIT 100` is global, not a quota per motif or size. It does not specify random sampling, seeds, candidate expansion, type/name acceptance filters, or the complete archived size grid. The presence of cycle/clique branches defines queries, not observed cycle/clique outcomes in the seven-family study. No result from executing these queries is claimed. These boundaries prevent equating the supplied query with a fully reconstructed historical pipeline. Cypher syntax is documented by Neo4j [7].

```
CALL {
// Chain, four nodes.
    MATCH (a)-[r1]->(b)-[r2]->(c)-[r3]->(d)
    RETURN 'chain' AS motif, [a,b,c,d] AS ns, [r1,r2,r3] AS rs
    UNION ALL
// Outward star, four nodes.
    MATCH (h)-[r1]->(a), (h)-[r2]->(b), (h)-[r3]->(c)
    WHERE elementId(a) < elementId(b) AND elementId(b) < elementId(c)

    RETURN 'star_out' AS motif, [h,a,b,c] AS ns, [r1,r2,r3] AS rs
    UNION ALL
// Inward star, four nodes.
    MATCH (a)-[r1]->(h), (b)-[r2]->(h), (c)-[r3]->(h)
    WHERE elementId(a) < elementId(b) AND elementId(b) < elementId(c)

    RETURN 'star_in' AS motif, [h,a,b,c] AS ns, [r1,r2,r3] AS rs
    UNION ALL
// Outward tree, five nodes.
    MATCH (h)-[r1]->(a)-[r3]->(c), (h)-[r2]->(b)-[r4]->(d)
    WHERE elementId(a) < elementId(b)
    RETURN 'tree_out' AS motif, [h,a,b,c,d] AS ns, [r1,r2,r3,r4] AS
      rs
    UNION ALL
// Inward tree, five nodes.
    MATCH (c)-[r3]->(a)-[r1]->(h), (d)-[r4]->(b)-[r2]->(h)
    WHERE elementId(a) < elementId(b)
    RETURN 'tree_in' AS motif, [h,a,b,c,d] AS ns, [r1,r2,r3,r4] AS
      rs
    UNION ALL
// Rhombus, four nodes.
    MATCH (a)-[r1]->(b)-[r3]->(d), (a)-[r2]->(c)-[r4]->(d)
    WHERE elementId(b) < elementId(c)
    RETURN 'rhombus' AS motif, [a,b,c,d] AS ns, [r1,r2,r3,r4] AS rs
    UNION ALL
// Hourglass, five nodes.
    MATCH (a)-[r1]->(m), (b)-[r2]->(m), (m)-[r3]->(c), (m)-[r4]->(d)
    WHERE elementId(a) < elementId(b) AND elementId(c) < elementId(d)

    RETURN 'hourglass' AS motif, [m,a,b,c,d] AS ns, [r1,r2,r3,r4] AS
      rs
    UNION ALL
// Directed cycle, three nodes.
    MATCH (a)-[r1]->(b)-[r2]->(c)-[r3]->(a)
    WHERE elementId(a) < elementId(b) AND elementId(a) < elementId(c)

    RETURN 'cycle' AS motif, [a,b,c] AS ns, [r1,r2,r3] AS rs
    UNION ALL
// Bidirected clique, three nodes.
    MATCH (a)-[r1]->(b), (b)-[r2]->(a), (a)-[r3]->(c), (c)-[r4]->(a),

          (b)-[r5]->(c), (c)-[r6]->(b)
    WHERE elementId(a) < elementId(b) AND elementId(b) < elementId(c)

    RETURN 'clique' AS motif, [a,b,c] AS ns, [r1,r2,r3,r4,r5,r6] AS
      rs
}
// Require distinct nodes.
WITH motif, ns, rs
WHERE all(n IN ns WHERE single(x IN ns WHERE x = n))
// Represent edges as source, type, target triples.
WITH motif, ns,
     [r IN rs | [elementId(startNode(r)), type(r), elementId(endNode
      (r))]] AS edges
// Reject any additional typed induced edge,
// including self-loops.
WHERE NOT EXISTS {
    MATCH (u)-[extra]->(v)
    WHERE u IN ns AND v IN ns
      AND NOT ([elementId(u), type(extra), elementId(v)] IN edges)
}
RETURN DISTINCT motif,
       [n IN ns | elementId(n)] AS node_ids,
       [n IN ns | coalesce(n.name, n.headline, n.title, n.ticker)]
      AS node_names,
       edges
LIMIT 100;
```

The supplementary files include the existing queries for exporting nodes and edges and separately labelled, newly written frozen-instance checks. These supplementary files were not used to generate the reported outcomes. Graph-export code is not itself historical motif-extraction evidence. Query IDs and role bindings refer to the same database snapshot; `elementId` is not a portable cross-snapshot identifier.